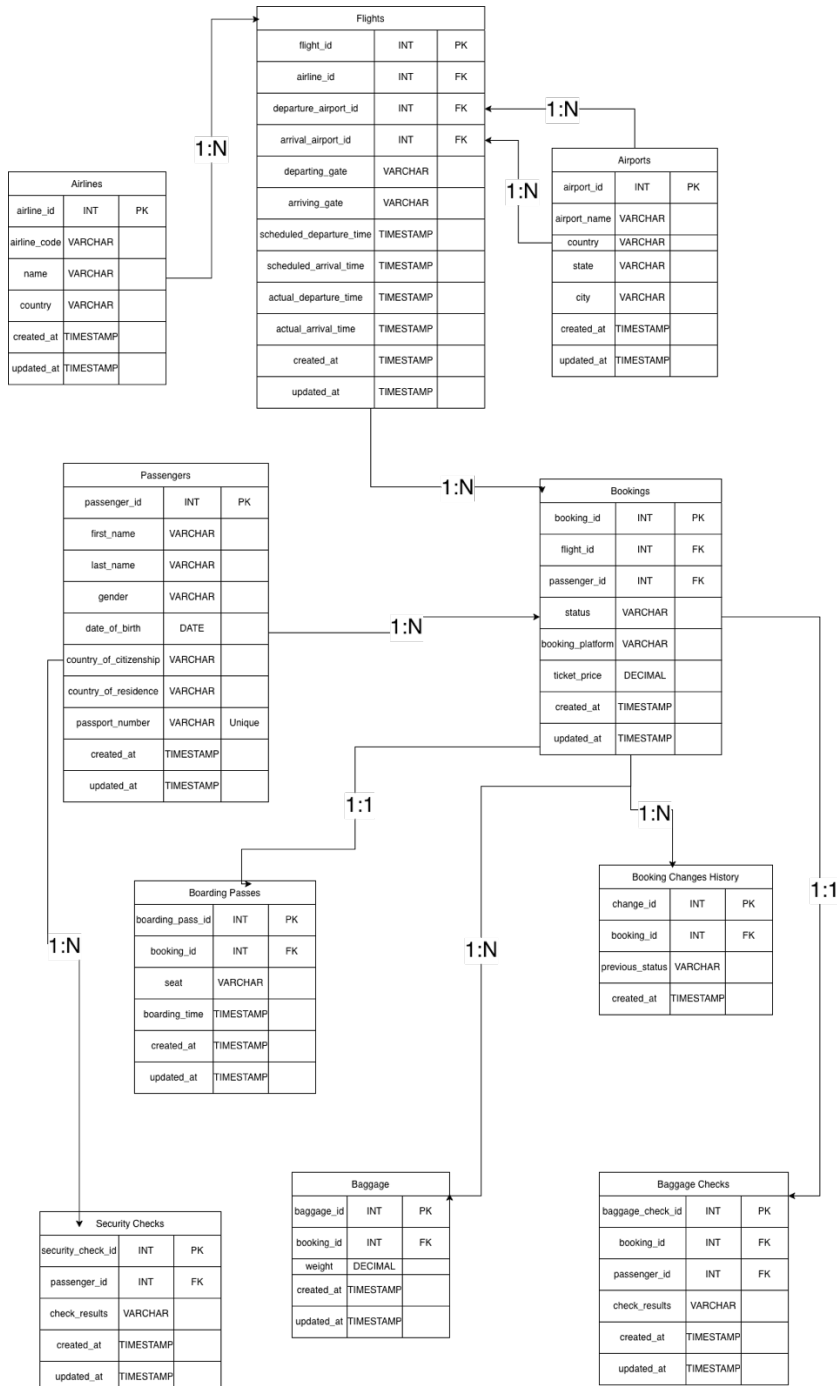


DATABASES

Laboratory Work 1: ERD Diagram

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1. System Overview

The International Airport Database management system is designed to handle core operations within a modern airport ecosystem. The primary objective of this database structure is to support passenger movement, flight scheduling, ticket reservations, security procedures, and baggage handling. By maintaining relational integrity across entities, the system ensures real-time tracking, seamless operational workflows, and data consistency from initial flight booking to final arrival and security clearance.

2. Notation Legend

To interpret the ER-Diagram accurately, the following symbols, keys, and relational constraints are defined:

- **PK (Primary Key):** An attribute or set of attributes that uniquely identifies each record in a entity table. PKs enforce entity integrity and prevent duplicate records.
- **FK (Foreign Key):** An attribute that creates a logical link between two entities by referencing the Primary Key of another table. FKs enforce referential integrity.
- **1:N (One-to-Many):** A cardinality relationship where a single record in the parent entity can be associated with multiple records in the related child entity, but a child record belongs to only one parent record.
- **1:1 (One-to-One):** A cardinality relationship where a single record in the first entity is strictly linked to at most one record in the second entity, and vice versa.

3. Comprehensive Entity & Attribute Description

• Airports

The **Airports** entity serves as a master dictionary for all origin and destination airports involved in flight routes. It allows the system to organize regional and international air traffic.

- **airport_id (INT, PK):** Unique numeric identifier assigned to each airport.
- **airport_name (VARCHAR):** Official name of the airport (e.g., Almaty International Airport).
- **country (VARCHAR):** Country where the airport operates.
- **state (VARCHAR):** State, province, or region where applicable.
- **city (VARCHAR):** Metropolitan area or city served by the airport.
- **created_at (TIMESTAMP):** System timestamp indicating when the airport record was originally entered.
- **updated_at (TIMESTAMP):** Timestamp indicating the most recent update to the airport's profile.

• Airlines

The **Airlines** entity stores registration details and identifiers of air carriers authorized to operate flights within the airport network.

- **airline_id (INT, PK):** Unique identifier for each airline company.
- **airline_code (VARCHAR):** Standard IATA/ICAO airline designator code (e.g., KC, DV, TK).
- **name (VARCHAR):** Commercial brand or legal name of the airline.
- **country (VARCHAR):** Country of primary registration/headquarters.
- **created_at (TIMESTAMP):** System creation timestamp.
- **updated_at (TIMESTAMP):** System last update timestamp.

• Flights

The **Flights** entity is the central operational unit, mapping scheduling and real-time execution of air journeys between departure and arrival locations.

- **flight_id (INT, PK):** Unique numeric ID for a scheduled flight instance.
- **airline_id (INT, FK):** References **Airlines** (**airline_id**) to identify the operating carrier.
- **departure_airport_id (INT, FK):** References **Airports** (**airport_id**) as the origin location.
- **arrival_airport_id (INT, FK):** References **Airports** (**airport_id**) as the destination location.
- **departing_gate (VARCHAR):** Terminal gate designated for passenger boarding.
- **arriving_gate (VARCHAR):** Designated arrival/disembarkation gate.
- **scheduled_departure_time (TIMESTAMP):** Published timetable departure time.
- **scheduled_arrival_time (TIMESTAMP):** Published timetable arrival time.
- **actual_departure_time (TIMESTAMP):** Actual time aircraft pushed back or took off.
- **actual_arrival_time (TIMESTAMP):** Actual time aircraft landed or arrived at the gate.
- **created_at (TIMESTAMP):** Timestamp when the flight schedule was added.
- **updated_at (TIMESTAMP):** Timestamp of any status/gate changes.

• Passengers

The **Passengers** entity maintains individual passenger profile records and demographic information required for identification and border clearance.

- **passenger_id (INT, PK):** Unique identifier assigned to each registered passenger profile.
- **first_name (VARCHAR):** Legal given name(s) matching identity documents.
- **last_name (VARCHAR):** Legal family name matching identity documents.
- **gender (VARCHAR):** Gender identity classification for travel security compliance.
- **date_of_birth (DATE):** Passenger date of birth for age verification and ticketing rules.
- **country_of_citizenship (VARCHAR):** Primary nationality of the passport holder.
- **country_of_residence (VARCHAR):** Primary country where the passenger resides.
- **passport_number (VARCHAR, Unique):** Official passport document number, required to be unique across profiles.
- **created_at (TIMESTAMP):** Profile creation timestamp.
- **updated_at (TIMESTAMP):** Last modification timestamp of personal details.

• Bookings

The **Bookings** entity represents transactional reservations made by passengers for specific flights.

- **booking_id (INT, PK):** Unique reservation reference ID.
- **flight_id (INT, FK):** References **Flights** (**flight_id**) indicating the booked flight segment.
- **passenger_id (INT, FK):** References **Passengers** (**passenger_id**) identifying the traveler.
- **status (VARCHAR):** Current status of the reservation (e.g., Confirmed, Cancelled, Pending).
- **booking_platform (VARCHAR):** Distribution channel used for purchase (e.g., Web Portal, Mobile App, Travel Agency).

- `ticket_price` (**DECIMAL**): Total financial amount paid for the ticket.
 - `created_at` (**TIMESTAMP**): Timestamp when reservation was created.
 - `updated_at` (**TIMESTAMP**): Timestamp when booking details/status changed.
- **Booking Changes History**
 This auditing entity records historical modifications made to reservations over time, ensuring traceability for ticket alterations or rebookings.
- `change_id` (**INT, PK**): Unique tracking ID for each logged modification event.
 - `booking_id` (**INT, FK**): References `Bookings` (`booking_id`) linking to the affected reservation.
 - `previous_status` (**VARCHAR**): Prior state or field value before the update occurred.
 - `created_at` (**TIMESTAMP**): Date and time when the change event occurred.
- **Boarding Passes**
 The `Boarding Passes` entity logs official clearance documents issued to passengers upon successful check-in, granting physical access to aircraft.
- `boarding_pass_id` (**INT, PK**): Unique identifier for the issued pass.
 - `booking_id` (**INT, FK**): References `Bookings` (`booking_id`) to link the pass with a confirmed ticket.
 - `seat` (**VARCHAR**): Assigned cabin seat designation (e.g., 12A, 04F).
 - `boarding_time` (**TIMESTAMP**): Scheduled time for passenger terminal boarding.
 - `created_at` (**TIMESTAMP**): Issuance timestamp.
 - `updated_at` (**TIMESTAMP**): Timestamp of seat changes or re-issuance.
- **Baggage**
 The `Baggage` entity tracks registered physical luggage checked in under a specific flight reservation.
- `baggage_id` (**INT, PK**): Unique identifier for each checked baggage item.
 - `booking_id` (**INT, FK**): References `Bookings` (`booking_id`) to attribute ownership to a reservation.
 - `weight_in_kg` (**DECIMAL**): Measured weight of the luggage piece in kilograms.
 - `created_at` (**TIMESTAMP**): Timestamp when luggage was logged at check-in.
 - `updated_at` (**TIMESTAMP**): Timestamp of any weight or status updates.
- **Baggage Checks**
 This entity logs safety screening and security inspection results performed on checked baggage prior to loading.
- `baggage_check_id` (**INT, PK**): Unique log ID for the baggage inspection instance.
 - `booking_id` (**INT, FK**): References `Bookings` (`booking_id`) to associate the inspection with the itinerary.
 - `passenger_id` (**INT, FK**): References `Passengers` (`passenger_id`) linking the inspected items directly to the traveler.
 - `check_results` (**VARCHAR**): Recorded inspection outcome (e.g., Cleared, Flagged, Manual Inspection Required).
 - `created_at` (**TIMESTAMP**): Inspection execution timestamp.
 - `updated_at` (**TIMESTAMP**): Last modification timestamp of check records.
- **Security Checks**
 The `Security Checks` entity records airport terminal checkpoint screenings for passengers prior to entering airside zones.
- `security_check_id` (**INT, PK**): Unique record ID for the passenger security event.
 - `passenger_id` (**INT, FK**): References `Passengers` (`passenger_id`) identifying the screened traveler.
 - `check_results` (**VARCHAR**): Result of personal screening (e.g., Passed, Secondary Screening Required).
 - `created_at` (**TIMESTAMP**): Timestamp when screening was performed.
 - `updated_at` (**TIMESTAMP**): Timestamp of record entry updates.

4. Relationships and Cardinality Matrix

Parent Entity	Child Entity	Relationship Type	Foreign Key(s)	Description / Business Rules
Airlines	Flights	1:N	airline_id	An airline can schedule and operate multiple flights, but each flight belongs to exactly one operating airline.
Airports	Flights (Departure)	1:N	departure_airport_id	An airport serves as the departure origin for many flights, while a flight departs from one origin airport.

Parent Entity	Child Entity	Relationship Type	Foreign Key(s)	Description / Business Rules
Airports	Flights (Arrival)	1:N	arrival_airport_id	An airport serves as the arrival destination for many flights, while a flight arrives at one destination airport.
Flights	Bookings	1:N	flight_id	A single flight holds multiple passenger bookings, whereas each booking corresponds to one specific flight segment.
Passengers	Bookings	1:N	passenger_id	A passenger can make multiple flight bookings over time, but each booking record is assigned to one primary passenger profile.
Bookings	Booking Changes History	1:N	booking_id	A booking can undergo several updates/modifications over its lifecycle, producing multiple history log entries.
Bookings	Boarding Passes	1:1	booking_id	Each confirmed booking yields exactly one boarding pass upon check-in for the scheduled flight.
Bookings	Baggage	1:N	booking_id	One booking allows a passenger to register and check in multiple pieces of luggage.
Bookings	Baggage Checks	1:N	booking_id, passenger_id	Registered luggage associated with a booking undergoes security checks, resulting in inspection records.
Passengers	Security Checks	1:N	passenger_id	A passenger undergoes terminal security screening for each journey, generating multiple screening records over time.

5. Conclusion

In this laboratory work, a comprehensive conceptual database model for an International Airport System was successfully designed, normalized, and documented. The resulting ER-Diagram maps out the core operational domains of airport management, cleanly handling passenger information, reservations, flight logistics, and multi-tier security screenings.

The structural design incorporates relational constraints (Primary and Foreign Keys) alongside defined cardinalities (One-to-Many and One-to-One) to guarantee referential integrity and eliminate redundancy. By utilizing structured timestamping across entities, the database ensures full auditability of operations. This conceptual ER-Diagram provides a scalable foundation for transitioning into physical database schema implementation, SQL table creation, and relational database queries.